QUASI-COMPACTNESS OF THE DIAGONAL DESCENDS ALONG A GERBE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that the structural morphism of a gerbe onto its base induces a surjective morphism from the source to the square of the base. Invoked inside the counted proof (source lines 6666-6668). Supplies surjectivity of the arrow in the displayed diagram.
- The statement that in a 2-commutative triangle with a surjective arrow, quasi-compactness of one leg forces quasi-compactness of the other. Invoked inside the counted proof (source lines 6669-6671). Deduces quasi-compactness of the arrow into the square of the base from quasi-compactness of the composite.
- The descent statement for quasi-compactness along a flat cover of finite presentation. Invoked at the end of the counted proof (source lines 6672-6676). Concludes quasi-compactness of the diagonal of the base from the cartesian square.

Source reference. Original lemma lemma-descent-quasi-compact: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6676–6676. Navigation pointer only: the counted text refers to the Stacks item lemma-descent-quasi-compact, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-surjection-from-quasi-compact: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6673–6673. Navigation pointer only: the counted text refers to the Stacks item lemma-surjection-from-quasi-compact, which belongs to another chapter of the project and is not

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Source reference. Original proposition proposition-when-gerbe-over: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6669–6669. Navigation pointer only: the counted text refers to the Stacks item proposition-when-gerbe-over, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks such that $\mathcal{X}$ is a gerbe over $\mathcal{Y}$. If $\Delta_{\mathcal{X}}$ is quasi-compact, so is $\Delta_{\mathcal{Y}}$.*

Proof. Consider the diagram

$$\begin{array}{ccccc} \mathcal{X} & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} & \longrightarrow & \mathcal{X} \times \mathcal{X} \\ & & \downarrow & & \downarrow \\ & & \mathcal{Y} & \longrightarrow & \mathcal{Y} \times \mathcal{Y} \end{array}$$

By Proposition proposition-when-gerbe-over we find that the arrow on the top left is surjective. Since the composition of the top horizontal arrows is quasi-compact, we conclude that the top right arrow is quasi-compact by Lemma lemma-surjection-from-quasi-compact. The square is cartesian and the right vertical arrow is surjective, flat, and locally of finite presentation. Thus we conclude by Lemma lemma-descent-quasi-compact. □